



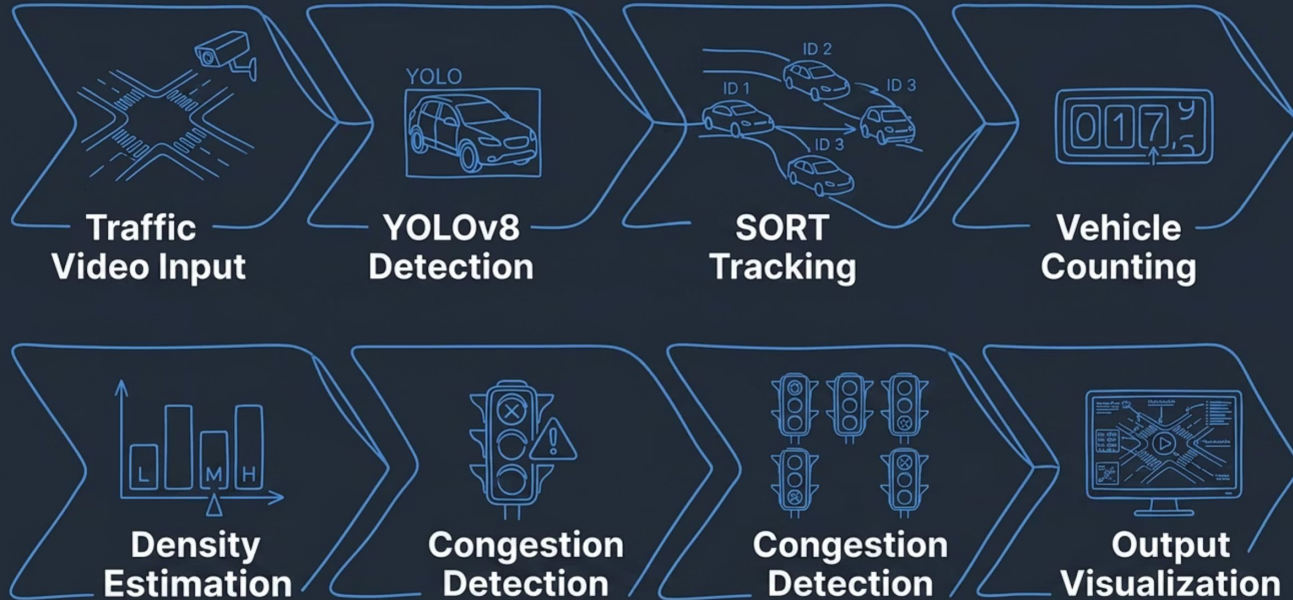
INTELLIGENT TRAFFIC SYSTEMS

Implementation Methodology

An intelligent traffic monitoring and adaptive signal control system using **Python**, **OpenCV**, **YOLOv8**, and **SORT tracking** — processing intersection video to detect vehicles, estimate traffic density, generate congestion warnings, and dynamically allocate adaptive green signal timing.

Methodology Workflow

The system follows a structured pipeline from raw video input to intelligent signal output, with each stage building on the previous to enable real-time adaptive traffic control.



Traffic Video Input

Input Processing

A traffic intersection video was provided as input. The video was processed frame-by-frame using OpenCV. Each frame was analyzed for vehicle detection and tracking.

Technologies Used

- Python
- OpenCV

Processing Flow

Each frame of the intersection video is loaded sequentially and preprocessed before being passed to the YOLOv8 detection model for analysis.

Vehicle Detection using YOLOv8

What is YOLOv8?

YOLOv8 (You Only Look Once) was used for real-time object detection. Each frame was passed to the YOLOv8 model. Bounding boxes were generated around detected vehicles. Low-confidence detections were filtered using a confidence threshold.

Detected Vehicle Types

- Cars
- Motorcycles
- Buses
- Trucks

Advantages

- Fast real-time detection
- High detection accuracy
- Suitable for intelligent traffic systems

Vehicle Tracking using SORT

SORT (Simple Online Realtime Tracking) was used for multi-object tracking. Unique IDs were assigned to vehicles across consecutive frames, ensuring each vehicle received a unique tracking ID and vehicle movement was tracked continuously.



Unique ID Assignment

Each vehicle received a unique tracking ID maintained across consecutive frames, enabling consistent monitoring throughout the video.



Prevent Duplicate Counting

Tracking reduced duplicate counting and improved traffic monitoring reliability by maintaining vehicle identity across frames.



Improve Density Estimation

Continuous tracking of vehicle movement improved the accuracy of traffic density estimation across frames.

Vehicle Counting, Density Estimation & Congestion Warning

Density Classification

Vehicle count was estimated using tracked vehicles. Traffic density levels were classified into Low, Medium, and High based on the total number of vehicles detected per frame.

Vehicle Count	Density Level
< 3	Low Density
3-5	Medium Density
≥ 6	High Density

Congestion Warning System

A congestion warning was generated when vehicle count exceeded the predefined threshold.



Congestion Threshold: Vehicle Count > 5

Output Warning: CONGESTION ALERT!

Adaptive Traffic Signal Timing

The system dynamically adjusted green signal timing according to traffic density, responding to congestion rather than using a fixed schedule.

Signal Allocation Logic

$$\text{Adaptive Green Time} = \text{MIN_GREEN} + (\text{Vehicle Count} \times 2)$$

Signal Timing Constraints

- Minimum Green Time = 10 seconds
- Maximum Green Time = 60 seconds

Fixed-Time vs. Adaptive Signals

Parameter	Fixed-Time	Adaptive
Green Duration	Constant	Dynamic
Traffic Adaptation	None	Responds to congestion
Peak Traffic Efficiency	Inefficient	Better traffic flow
Waiting Time	Higher	Reduced

Demo of the Developed Code

The demonstration walks through all seven stages of the system in action, from loading the intersection video to generating the final annotated output.

01

Input Traffic Video

A traffic intersection video was loaded into the system. The video contained multiple moving vehicles in urban traffic conditions.

02

Vehicle Detection

YOLOv8 detected vehicles in each frame. Bounding boxes were drawn around detected vehicles: cars, motorcycles, buses, and trucks.

03

Vehicle Tracking

SORT tracking assigned unique IDs to detected vehicles (e.g., ID: 1, ID: 2, ID: 3). Tracking ensured consistent monitoring across frames.

04

Counting & Density Estimation

Vehicle count was calculated for each frame. Traffic density was estimated based on the total number of vehicles. Example output: Vehicles: 9, Density: High.

05

Congestion Warning Display

When traffic density exceeded the threshold, the system displayed: CONGESTION ALERT! This indicated heavy traffic congestion at the intersection.

06

Adaptive Signal Timing

Green signal timing was adjusted dynamically according to vehicle density. Example: Fixed Green: 20s → Adaptive Green: 34s. Higher traffic density received longer green duration.

07

Output Video Generation

The processed frames were saved into an output video displaying vehicle detection, tracking IDs, density information, congestion alerts, and adaptive signal timing.

Output Observations & Findings

Vehicle Detection Performance

YOLOv8 successfully detected multiple vehicle categories in real-time traffic video. Detection performance was accurate for most vehicles in normal lighting conditions. Bounding boxes clearly identified vehicles in each frame.

Vehicle Tracking Performance

SORT tracking maintained unique IDs for vehicles across frames. Tracking reduced duplicate counting and improved traffic monitoring reliability.

Traffic Density Estimation

The system successfully classified traffic conditions into low, medium, and high density. Traffic density increased significantly during crowded traffic scenes.

Congestion Detection

Congestion alerts were generated whenever vehicle count exceeded the threshold. The warning system effectively identified highly congested traffic conditions.

Adaptive Signal Timing Performance

Adaptive signal timing dynamically increased green signal duration during heavy traffic conditions. During low-density traffic, green signal duration remained shorter, reducing unnecessary waiting.

Comparative Analysis & Limitations

Fixed-Time vs. Adaptive Signal Analysis

Parameter	Fixed-Time	Adaptive
Responsiveness	Static	Dynamic
Traffic Adaptation	Poor	Better
Waiting Time	Higher	Reduced
Congestion Handling	Inefficient	Improved
Traffic Flow Efficiency	Moderate	Higher

Current Limitations

- The current implementation supports only a single traffic intersection.
- Reinforcement learning and swarm intelligence optimization are not yet implemented.
- Detection accuracy may decrease in highly crowded or low-visibility environments.